Data

Assessment data

Objective Knowledge

Students completed short, multiple-choice measures of objective knowledge before the pre-lab and at the end of the complete lab. Items on the objective knowledge test were matched to learning objectives and skills for the activity (items are listed below).

Examining scores overall showed large gains in knowledge related to the learning objectives:

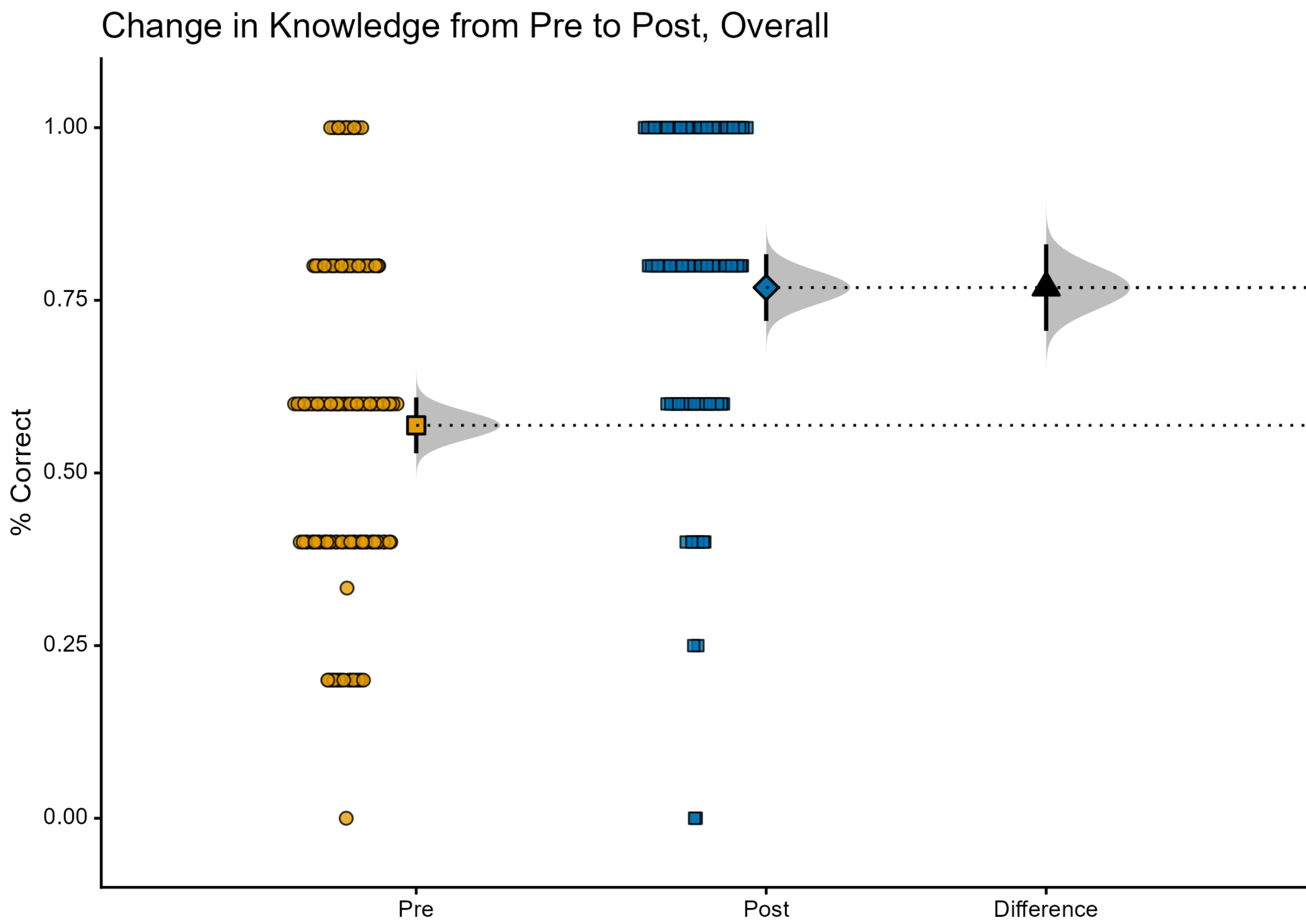

Figure 1: Changes in Objective Knowledge Scores; Individual data, means, and 95% CIs

Effect	Effect Size	LL	UL	SE	N
Post	0.76	0.72	0.81	0.02	130
Pre	0.56	0.52	0.60	0.02	91
Post-Pre	0.19	0.13	0.26	0.03	

In standardized terms: d_{unbiased} = 0.85 95% CI [0.57, 1.13]

Breaking this down by school shows some diversity in student learning, with students completing the activity in an online setting (Dominican University) showing the smallest gains:

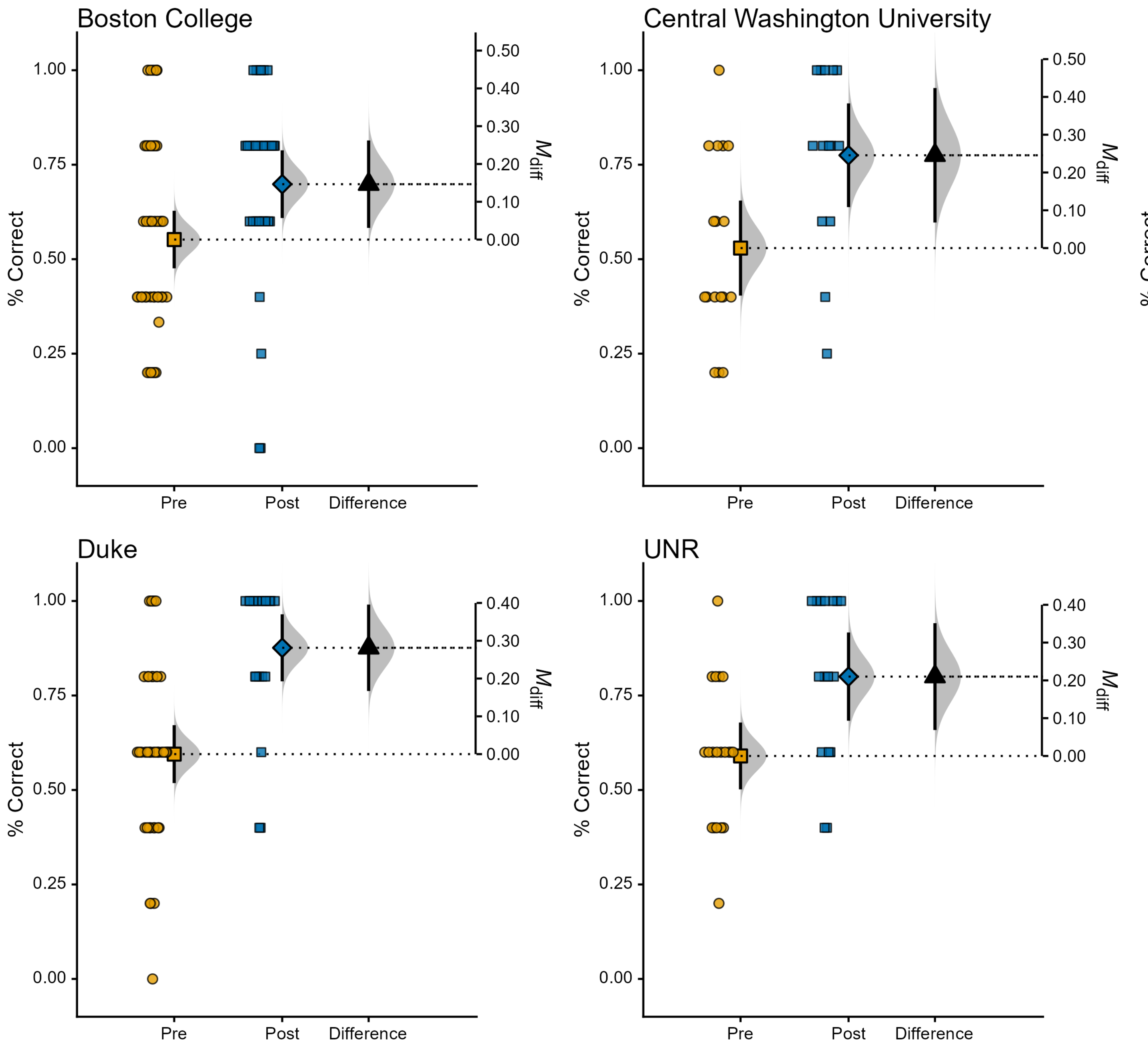

Figure 2: Objective Learning Gains by School; Individual data, means, and 95% CIs

Confidence Ratings

Students rated their confidence on knowledge and skills related to the activity both before the pre-lab and after completing the lab. Confidence was rated on a scale from 1-5. Self-reported learning gains have poor validity, but we did not very large increases in student confidence for knowledge and skills related to the activity.

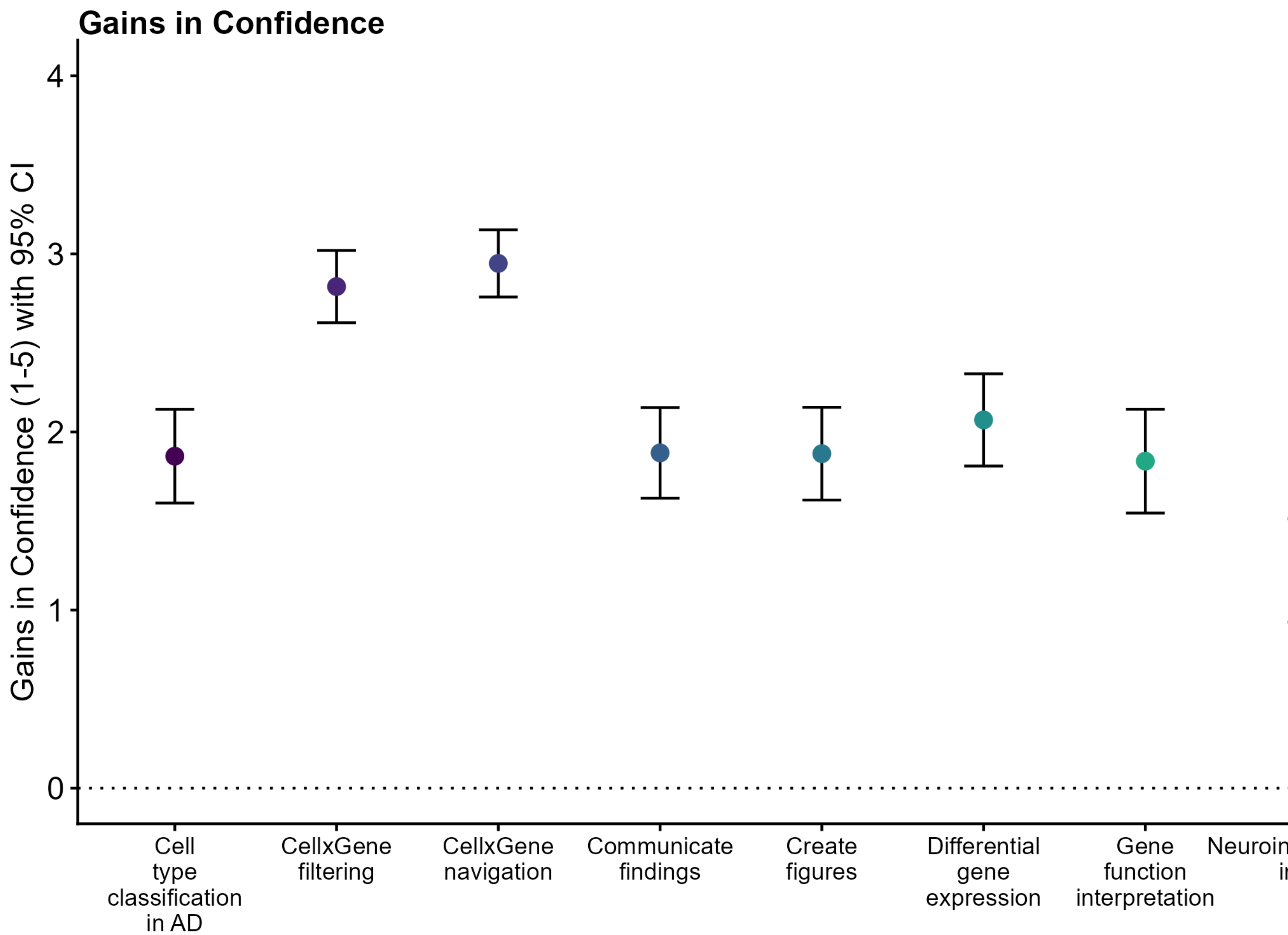

Figure 3: Gains in Confidence Ratings (Pre to Post, 1-5 scale); Dotted-line at 0 indicates no gains; Means and 95% CIs

Confidence item	Effect Size	LL	UL	SE	df
RNA-seq principles	1.16	0.88	1.44	0.14	222.79
Neuroinflammation in AD	1.22	0.93	1.51	0.15	223.73
Cell type classification in AD	1.86	1.60	2.13	0.13	216.89
UMAP interpretation	2.83	2.61	3.06	0.11	199.15
CellxGene navigation	2.95	2.76	3.14	0.10	212.59
Qualitative gene expression comparison	2.62	2.39	2.86	0.12	224.95

Confidence item	Effect Size	LL	UL	SE	df
CellxGene filtering	2.82	2.61	3.02	0.10	217.11
Differential gene expression	2.07	1.81	2.33	0.13	222.93
Create figures	1.88	1.62	2.14	0.13	211.24
Gene function interpretation	1.84	1.54	2.13	0.15	226.00
Communicate findings	1.88	1.63	2.14	0.13	225.21

Breaking this down by school showed roughly similar levels of confidence gains across contexts:

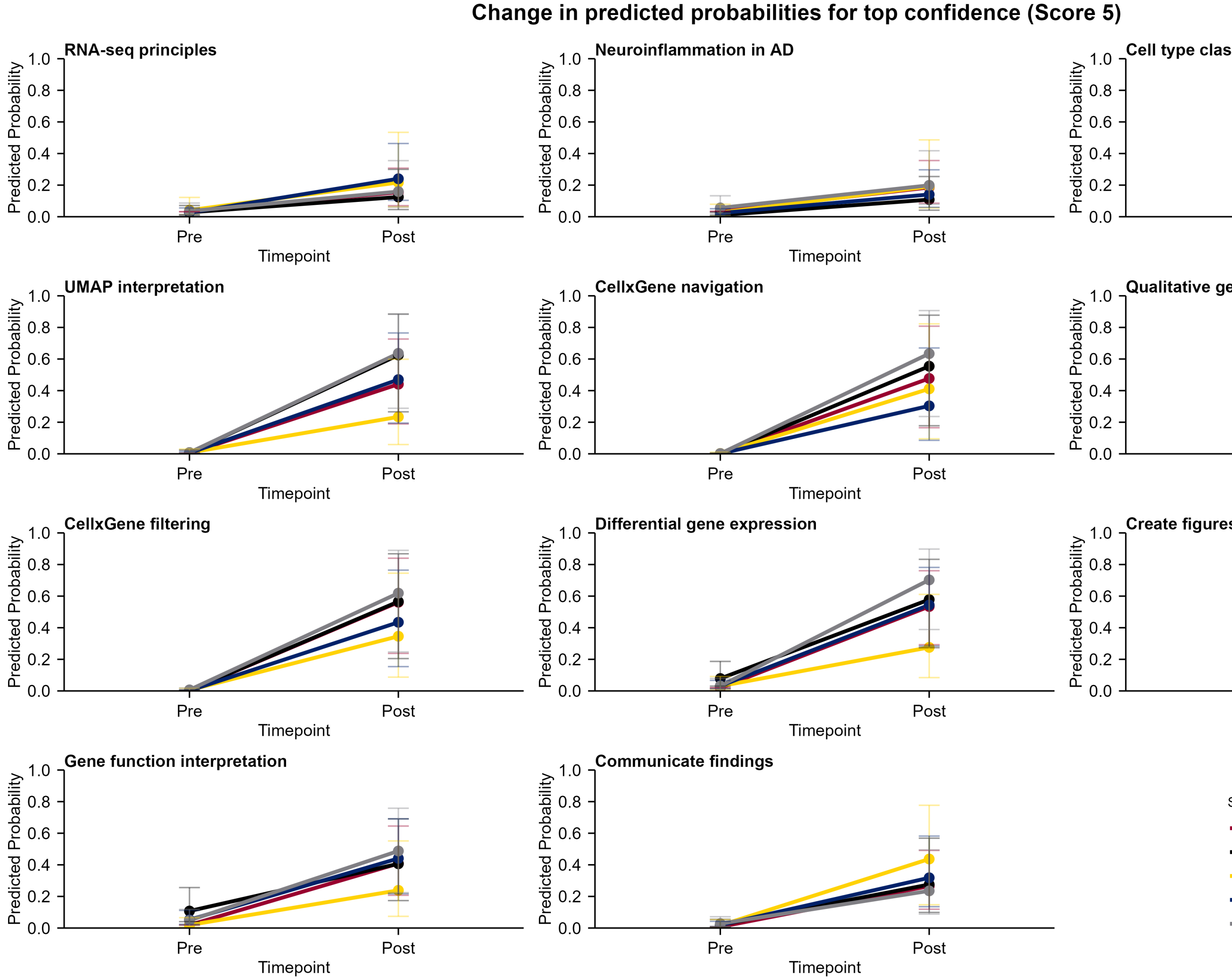

Figure 4: Changes in Chance of Highest Confidence by Item and School; Modeled means and CIs

Feedback

Finally, students gave feedback on different aspects of the activity (1-5 rating scale). Most students gave most aspects of the lab strong feedback. Note that lower scores on "Preferred lecture format".

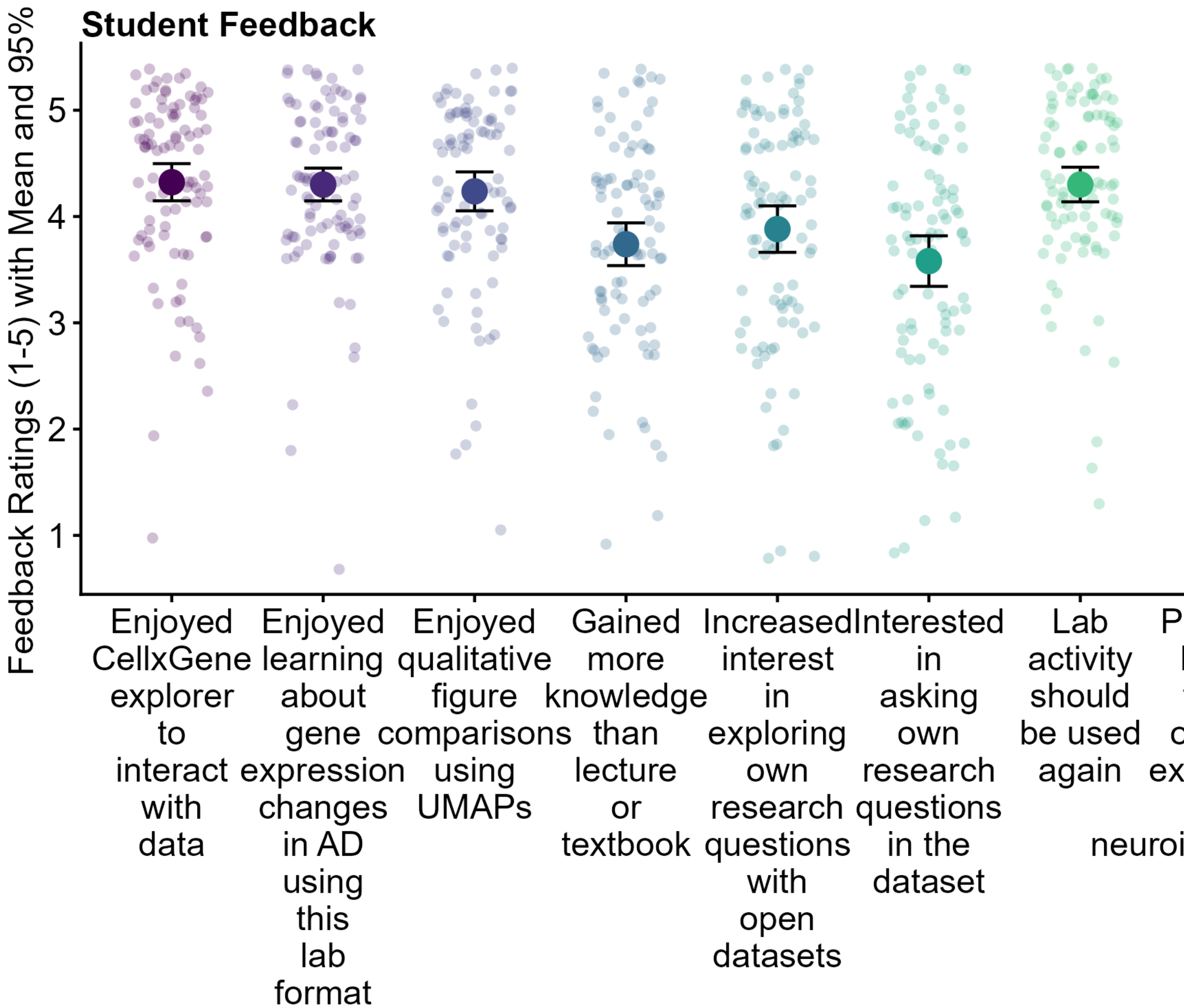

Figure 5: Student Feedback on Aspects of the Lab; Individual data, means, and 95% CIs

Feedback Item	Effect Size	LL	UL	SE	df
Enjoyed learning about gene expression changes in AD using this lab format	4.30	4.15	4.46	0.08	92
Preferred lecture format on gene expression and neuroinflammation in AD	2.84	2.56	3.11	0.14	92
Gained more knowledge than lecture or textbook	3.74	3.54	3.94	0.10	91
Enjoyed CellxGene explorer to interact with data	4.32	4.15	4.50	0.09	92
Enjoyed qualitative figure comparisons using UMAPs	4.24	4.05	4.42	0.09	92
Lab activity should be used again	4.30	4.14	4.46	0.08	92
SEA-AD helped me understand real-world data used to study AD	4.37	4.19	4.54	0.09	92

Feedback Item	Effect Size	LL	UL	SE	df
Interested in asking own research questions in the dataset	3.58	3.34	3.82	0.12	92
Increased interest in exploring own research questions with open datasets	3.88	3.66	4.10	0.11	92
See value in open datasets in future coursework and research	4.31	4.15	4.48	0.08	92